

Applying foundation model embeddings towards urban livability evaluation

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Abstract

While accurate measurement of socioeconomic indicators remains challenging in data-scarce regions, which limits policy interventions and resource allocation, high-resolution geospatial data is widely available and can contain information on various livability statistics. We investigate which physical features are encoded within foundation model embeddings, such as AlphaEarth, AnySat, and TerraMind, and provide a systematic framework for identifying the most predictive geospatial indicators. By analyzing how different types of geospatial data influence urban livability predictions, our approach enables researchers to prioritize the most informative features for their specific applications. Additionally, we demonstrate how to leverage foundation model embeddings to enhance prediction performance for these outcomes. This work contributes a principled methodology for extracting actionable information from satellite imagery while accounting for complex spatial dependencies, with applications in predicting urban livability in regions with limited observation data.

CCS Concepts

• **Computing methodologies** → **Scene understanding**; *Transfer learning*; • **Information systems** → **Geographic information systems**.

Keywords

AlphaEarth, TerraMind, AnySat, Geospatial Embeddings, Urban livability, Multimodal deep learning, Satellite images, Digital surface model, Nighttime light remote sensing, Textual information

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1 Introduction

Urbanization is reshaping the modern world. Nearly 45% of the world's 8.2 billion people currently live in cities, and this share is projected to reach 68% of the world's 9.7 billion people by 2050, with two-thirds of global growth projected to occur in cities within the next 25 years. As such, it is important to understand and improve the quality of life within cities. Although well-being is subjective, the Sustainable Development Goal 11 provides a shared framework, calling for cities and settlements to be inclusive, safe, resilient, and sustainable.

Previous research on assessing urban livability has focused on statistical methods to estimate livability metrics [18, 20, 25]. However, these methods are not generalizable as indicators are selected manually. The authors of [34] address this gap by using a variety of geospatial data for deep learning-based livability evaluation. These data sources are collected and processed independently, requiring significant effort to integrate. Furthermore, potentially valuable datasets may have been overlooked or excluded due to collection costs, time constraints, or subjective judgement. Geospatial embeddings offer a promising avenue to address these drawbacks. These foundational embeddings assimilate a diverse range of data sources, including optical, radar, environmental, and textual information into real-numbered vectors. This could reduce data engineering overhead while potentially improving livability predictions through unifying data modalities. We specifically examine AlphaEarth Foundations [5], TerraMind [13], and AnySat [3] to understand their impact on urban livability evaluation. We build on the Transformer from [34] to include additional embeddings and explore how these representations interact with different data modalities. The contribution of this paper lies not in extending the model to incorporate embeddings, but rather in analyzing how different geospatial embeddings, both individually and in combination, influence livability prediction. We also conduct interpretability analyses to better understand how these embeddings affect the model's internal representations and decision-making processes.

2 Related Work

2.1 Urban Livability

Previous research on assessing livability has focused on using statistical methods with chosen indicators. For example, 20 objective indicators were identified by statistical, remote sensing, and spatial



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data to assess the livability of urban areas using an analytic hierarchy process method [20]. The authors of [18] have focused on making livability multi-dimensional instead of over-emphasizing on economic aspects by relying on five major categories: education, medical services, recreation, transportation services, and living services. Recent work has used government data [21] and OpenStreetMap [22] to introduce a foundational livability indicator averaged across two domains: accessibility and economy. While these approaches offer valuable insights into livability, they often lack consistency between regions or fail to integrate diverse data sources. The Leefbaarometer project (LBM) [4] initiated by the Dutch government addresses these limitations by integrating a broad range of livability-influencing indicators into nationally consistent evaluation results, identifying 47 environmental characteristics linked to how much residents enjoy and value living in their areas. Although many geospatial datasets hold information relevant to livability [32], only 8.8% of the 68 studies reviewed in [14] make use of them. The authors of [34] address this underutilization through a transformer-based deep learning model to evaluate urban livability. They use a wide variety of large-volume data sources, which can be difficult to collect, preprocess, integrate, and train. Geospatial embeddings can serve as a gateway towards bridging this gap by offering precomputed, spatially grounded representations that consolidate diverse data sources without requiring task-specific collection and processing pipelines.

2.2 Geospatial Embeddings

In this work, we examine three different foundational embeddings: AlphaEarth, AnySat, and TerraMind. AlphaEarth Foundations [5] is an embedding field model that unifies a wide variety of Earth observation data. They use a transformer-based architecture to fuse the data sources into unit-normalized 64-dimensional vectors, generated at a 10-meter spatial resolution. These are already being used to enhance geospatial tasks, including data generation [10], retrieval-augmented generation [24], and height mapping [7]. AnySat [3] is a self-supervised geospatial model based on joint embedding predictive architecture and scale-adaptive spatial encoders. They use modalities with different resolutions to create 1536-dimensional vectors with a spatial resolution of 10m. The authors demonstrated its use on tasks like land cover mapping, deforestation detection, and flood segmentation [3]. Finally, TerraMind [13] is another multimodal foundation model that introduces a dual-scale encoder-decoder architecture combining token-level and pixel-level representations across many modalities. These 384-dimensional embeddings are offered at 160m spatial resolution, and they have been applied towards wildfire mapping [27] and flood susceptibility mapping [16].

3 Data

The Leefbaarometer (LBM) project [4] version 3.0 maintains national-level urban livability measurements across various cities in the Netherlands. They provide a wide range of spatial scales: grid, neighborhood, district, postcode, and municipality. To maintain consistency with raster data, we use the LBM's smallest spatial unit, the $100\text{m} \times 100\text{m}$ grid. The overall livability is split up into five dimensions: physical environment (PHY), housing availability

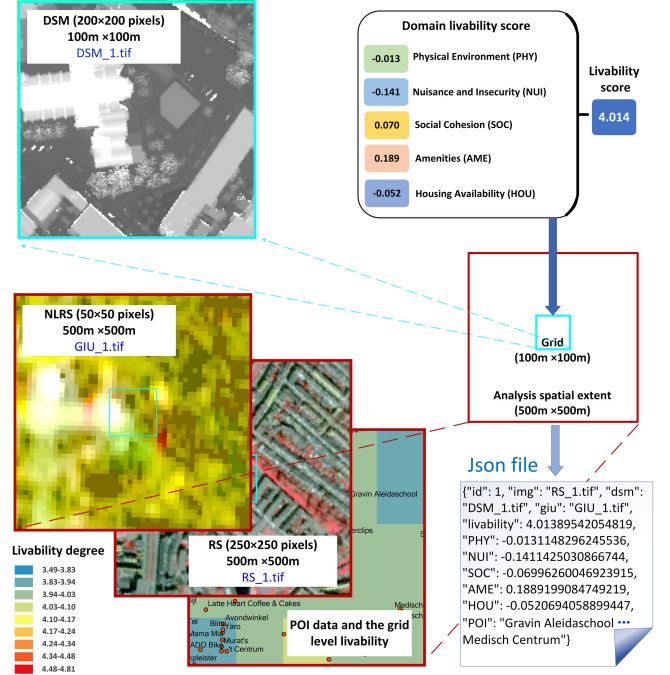


Figure 1: Overview of the dataset generation pipeline [34]. Remote sensing (RS), digital surface model (DSM), nightlight remote sensing (NLRS), and point-of-interest (POI) inputs are extracted at two spatial scales: DSM and livability scores cover the $100\text{m} \times 100\text{m}$ target grid, while RS, NLRS, and POI span a broader $500\text{m} \times 500\text{m}$ context window.

(HOU), amenities (AME), social cohesion (SOC), and nuisance and insecurity (NUI). The range of livability scores is from 3.3464 to 4.853, PHY from -0.142 to 0.118, NUI from -0.514 to 0.111, SOC from -0.151 to 0.095, HOU from -0.344 to 0.164, and AME from -0.080 to 0.695.

In the LBM project, the authors aggregate 47 environmental characteristics into these five domains [4, 34]. PHY is the broadest, spanning proximity to infrastructure such as major roads and high-voltage lines, natural amenities like green space and water, land-use mix, and risk factors including flooding, heat stress, and air quality. HOU is narrower, covering residential areas, housing vacancy, and the height, type, and age of buildings. AME reflects distance to and density of educational, healthcare, cultural, and retail facilities, along with transit and job accessibility. SOC combines perceived social cohesion with residential turnover, population density, household growth or decline, and age diversity. NUI captures reported violent crime, vandalism, and disorderly conduct, alongside residents' perceived disorder and fear of crime.

Following [34], we select the following input data: remote sensing (RS) images, digital surface model (DSM), nightlight remote sensing (NLRS) images and point-of-interest (POI) data. These modalities often have information that capture livability and its many domains. For example, RS data can infer car density information [1] (PHY), NLRS can infer population density [35] (SOC) and burglary [12] (NUI), DSM can infer building height (HOU), and

POI data can infer facility use (AME). No fixed set of modalities can exhaustively capture all factors that influence livability. As such, we also consider AlphaEarth, AnySat, and TerraMind embeddings to understand whether they can capture urban livability statistics. All data sources correspond to the year 2020. The RS images are sourced from SuperView multispectral satellite images at a 2m spatial resolution. The DSM is provided by the Dutch public geo-services platform (PDOK) with a 0.5m spatial resolution. The NLRs images are SDGSAT-1 Glimmer images at a 10m spatial resolution. All of the embeddings are accessible through the *rs-embed* tool [31].¹

Figure 1 illustrates how the dataset was constructed. We consider a buffer of 200m outside the evaluation grid cell, creating a total spatial extent of 500m. The DSM was confined within the grid while the RS, NLRs, POI, and embeddings included the buffer. As such, the pixel dimensions for RS, DSM, NLRs, AlphaEarth, and AnySat are respectively 250, 200, 50, 50, and 50. Because TerraMind uses a Vision Transformer with a fixed input resolution of 224×224 and patch size of 16, the 500 m window is resized to fit this input, producing pixel dimensions of 14. The number of samples for training, validation, and testing are respectively 29308, 9253, and 13440. More information on the dataset collection can be found in [34].

We adopt the training, validation, and testing split used by [34] rather than constructing our own. Out of the 13 Dutch cities in the dataset, four areas (Eindhoven, Hengelo, Dordrecht, and Beesel) are held out and used for testing. The remaining nine areas (Almere, Amsterdam, Arnhem, Eemsdelta, Groningen, Nijmegen, Rotterdam, Venlo, and Weert) are split across training and validation. As such, our reported test performance also measures generalization to entirely unseen cities rather than interpolating within a city. This setting matters as this is a motivating use case for using embeddings in new, unlabeled regions.

4 Model

Multimodal deep learning is promising in fusing features from different geospatial modalities [15, 23, 30]. In our work, we use three primary types of information: images, text, and embeddings. While there has been prior work on multimodal fusion in geospatial tasks [33], there has been little attention paid to integrating learned embeddings with raster imagery or text. To address this gap, we extend the transformer-based multi-task multimodal regression (TMTMR) model [34] to include additional embedding vectors.

When extending transformers to multimodal inputs or imagery, it is common to extract features from modalities before further processing [17, 33]. Pretrained models are exceptionally important, in this case, due to the lack of training samples.

For example, pretrained convolutional neural networks (CNNs) are often used as feature extractors for images. Some notable examples include ResNet [8], EfficientNetV2 [28], and DenseNet [11]. Of these, DenseNet is significant for its dense block architecture, allowing it to learn both local and global interactions in geospatial applications [33]. As illustrated in fig. 2, the images are normalized and resized using bilinear interpolation to match the input requirements of DenseNet. The images are then processed through the

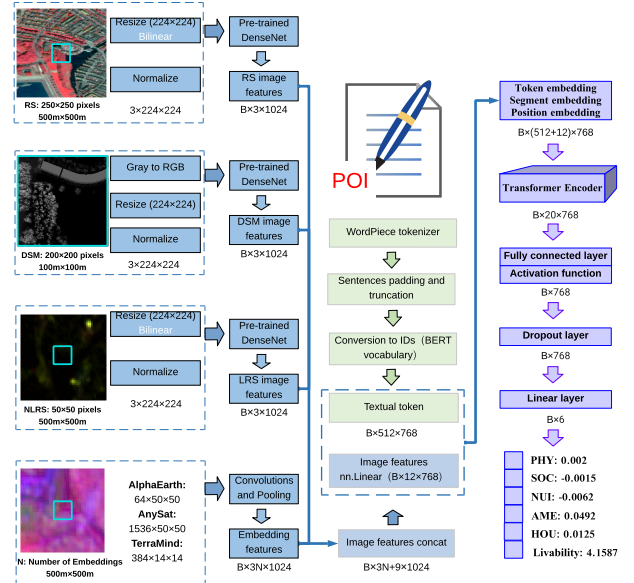


Figure 2: The architecture of the livability study regression model, inspired by [34]. Features of RS, DSM, NLRs are extracted by the pretrained DenseNet, features of embeddings are extracted by convolutional layers, and features of POI are extracted by the pretrained BERT network. These features are then concatenated and input into the Transformer Encoder, and finally decoded by a linear layer.

pretrained DenseNet to extract relevant spatial features. A powerful feature extractor for text is the Bidirectional Encoder Representation from Transformers (BERT) [6]: a pretrained language representation model that has been widely proficient in natural language processing tasks [2]. The words in the POI data are first tokenized into subwords and characters using the WordPiece tokenizer. Next, these tokens are truncated or padded to 512 tokens, which is the input size for BERT. Then, each token is mapped to its corresponding ID from the BERT vocabulary.

Geospatial embeddings, on the other hand, differ from standard images. Each pixel contains a high-dimensional vector rather than three channels. Because there are no suitable pretrained models, we utilize a dedicated series of convolutions and pooling layers to extract essential features from each embedding. We can append additional convolutional feature extractors to add more embeddings as inputs.

All image, embedding, and text features are then merged together before being put into a Transformer Encoder. A decoder comprising linear and dropout layers then outputs the overall livability score and five domain scores. The loss we use is a multitask MAE objective:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N \sum_{t=1}^T |\hat{y}_{i,t} - y_{i,t}| \quad (1)$$

¹<https://github.com/cybergis/rs-embed>

where N are the number of samples, T are the number of tasks, $\hat{y}_{i,t}$ is the predicted score of the i -th sample and $y_{i,t}$ is the true score of the i -th sample. This allows the model to learn shared features across tasks, harnessing the relationships between modalities. More information on the model and training process is available in Appendix A.

5 Experiments

Using the training dataset described in Section 3 and the model illustrated in Section 4, we conducted performance experiments on the effect of including geospatial embeddings. The baseline model (BASE) includes the original four modalities: RS, DSM, NLRs, and POI data. The augmented models (AEF, AS, TM) each include the previous four inputs as well as embeddings from AlphaEarth Foundations, AnySat, and TerraMind, respectively. In a few experiments, we include combinations of these embeddings as well to understand whether they provide complementary information or are redundant. Specifically, we evaluate all pairwise combinations (AEF+AS, AEF+TM, AS+TM) and the full combination (AEF+AS+TM) to assess the marginal contribution of each embedding source. Following standard foundation model literature [5], we include a baseline embedding-only linear probe for each foundation model (PROBE-AEF, PROBE-AS, PROBE-TM) to assess how much the multimodal architecture improves the performance. Each probe is trained on the spatially mean-pooled embedding vector and uses the same loss function. We test out different training input combinations and the model's robustness to missing inputs. To compare the performance, we report the RMSE of the six livability scores explained in Section 3.

We compare against the published TMTMR baseline in Appendix B, and BASE trails the published performance by roughly 5 to 11% RMSE across the six livability scores. We attribute this to training choices like epoch length and loss function rather than modeling decisions. Therefore, all embedding comparisons are relative to our own implementation rather than the published baseline. Replication code is available at <https://github.com/cybergis/embed-livability>.

5.1 Results on adding embeddings

As illustrated in Table 1, we compare the baseline with augmented models. Including AnySat in the model results in similar performance to the baseline model, while TerraMind provides a small but fairly consistent improvement across scores. The inclusion of AlphaEarth decreases RMSE in LIV and AME with slight improvements in other scores. This suggests that AlphaEarth's embeddings capture amenity-related information not encoded by AnySat or TerraMind. The AME RMSE further decreases when combining AlphaEarth with other embeddings. This may suggest that AnySat and TerraMind must capture some orthogonal signals that further improve AME performance when combined with AlphaEarth.

The embedding-only probes are used to understand how each embedding alone affects the performance. All probes underperform the BASE and single-embedding models on LIV and NUI, confirming that the multimodal architecture contribute a significant amount of information beyond the embeddings. SOC and HOU are closer with PROBE-TM having similar RMSE performance to BASE. AME and PHY

are interesting domains since PROBE-AEF beats the performance of BASE in both of these domains. This indicates that PHY- and AME-relevant information like proximity to infrastructure or amenity density is especially concentrated in these embeddings.

Although combining embeddings improves performance upon linear probes, we caution against over-interpreting this as clean evidence of complementary information. These embeddings are all pretrained on overlapping Earth observation sources, and our geospatial inputs are themselves drawn from related satellite and elevation products. A decrease in RMSE when including multiple embeddings could also be caused by added model capacity or each embedding contributing noisy versions of the same signal, which the model can ensemble into a better estimate. The latter explanation is more likely to be the case due to the performance of the probes. Although AEF is clearly stronger than TM, their respective embedding-only probes PROBE-AEF and PROBE-TM have very similar performance. If AlphaEarth's advantage in the multimodal model simply reflected carrying more raw livability signal, we would expect that to show up in the probe. Instead, the two embeddings are comparable alone but diverge once combined with geospatial inputs. These results are more consistent with AlphaEarth integrating synergistically with the other modalities rather than AlphaEarth have more information. Disentangling these explanations would require representation-level probing or using a formal complementary index [29] that is beyond the scope of this paper.

We split up the testing set into four cities with different traits: Eindhoven (a modern city), Hengelo (a slower-growing city), Dordrecht (a city with a well-preserved historical center), and Beesel (a rural community). The RMSE values for each area are summarized in Table 2. In urban areas (Eindhoven, Hengelo, and Dordrecht), including embeddings generally improves upon the baseline, with AlphaEarth usually providing consistent performance boosts across the livability scores. In Eindhoven, models that include AlphaEarth perform the best, with AEF and AEF+TM achieving the lowest LIV RMSE scores. In Hengelo and Dordrecht, there is no single embedding combination that consistently outperforms others across all scores. Notably, in Hengelo, the baseline achieves the best NUI RMSE, suggesting that embeddings do not always improve performance on all individual domains. The rural community of Beesel illustrates a significantly different pattern. Single embeddings, like AlphaEarth and AnySat, worsen LIV performance. Only combinations that pair AlphaEarth and TerraMind, such as AEF+AS+TM, recover and improve upon it. This implies that embeddings pretrained on predominantly urban geospatial data may not transfer cleanly to rural contexts. Including multiple embeddings may be necessary to mitigate this distributional mismatch.

In Table 3, we test out how well the baseline and embedding-augmented models perform in areas with and without text information. Consistent with Table 1, AEF+TM performs the best across both POI splits. All models perform worse with text information, which may be due to increased heterogeneity in dense areas with POI. However, the embeddings still significantly improve performance across the board, especially in areas without POI. In particular, AME RMSE improves from 0.048 to 0.040 with NULL POI, compared to 0.056 to 0.045 with ANY POI. This may reflect AlphaEarth's text-aware pretraining partially substituting for missing POI.

Table 1: RMSE comparison across livability domain scores between baseline network and models that include foundation model embeddings, and linear probes trained on pooled embedding vectors. Values are mean $\pm$ std over 3 runs. Best values for each score are bolded.

Model	LIV	PHY	NUI	SOC	AME	HOU
BASE	0.106 $\pm$ 0.002	0.027 $\pm$ 0.000	0.061 $\pm$ 0.001	0.032 $\pm$ 0.001	0.055 $\pm$ 0.002	0.034 $\pm$ 0.000
AEF	0.098$\pm$0.001	0.024$\pm$0.000	0.059$\pm$0.000	0.030$\pm$0.000	0.047 $\pm$ 0.001	0.033$\pm$0.000
AS	0.104 $\pm$ 0.000	0.028 $\pm$ 0.000	0.062 $\pm$ 0.001	0.032 $\pm$ 0.000	0.052 $\pm$ 0.001	0.034 $\pm$ 0.000
TM	0.101 $\pm$ 0.001	0.026 $\pm$ 0.000	0.060 $\pm$ 0.000	0.031 $\pm$ 0.000	0.052 $\pm$ 0.001	0.034 $\pm$ 0.000
AEF+AS	0.100 $\pm$ 0.001	0.025 $\pm$ 0.000	0.059$\pm$0.001	0.030$\pm$0.000	0.044$\pm$0.001	0.033$\pm$0.000
AEF+TM	0.098$\pm$0.000	0.025 $\pm$ 0.000	0.059$\pm$0.000	0.030$\pm$0.000	0.045 $\pm$ 0.001	0.033$\pm$0.001
AS+TM	0.101 $\pm$ 0.002	0.026 $\pm$ 0.000	0.060 $\pm$ 0.001	0.031 $\pm$ 0.001	0.050 $\pm$ 0.001	0.034 $\pm$ 0.000
AEF+AS+TM	0.098$\pm$0.001	0.024$\pm$0.000	0.059$\pm$0.000	0.030$\pm$0.000	0.044$\pm$0.001	0.033$\pm$0.001
PROBE-AEF	0.116 $\pm$ 0.002	0.024$\pm$0.001	0.064 $\pm$ 0.000	0.035 $\pm$ 0.000	0.053 $\pm$ 0.001	0.037 $\pm$ 0.000
PROBE-AS	0.130 $\pm$ 0.006	0.027 $\pm$ 0.001	0.071 $\pm$ 0.002	0.040 $\pm$ 0.002	0.060 $\pm$ 0.003	0.040 $\pm$ 0.002
PROBE-TM	0.114 $\pm$ 0.002	0.025 $\pm$ 0.001	0.064 $\pm$ 0.001	0.034 $\pm$ 0.000	0.056 $\pm$ 0.002	0.035 $\pm$ 0.001

Table 2: RMSE comparison across livability domain scores between baseline network and models that include foundation model embeddings. The city column indicates the city used for testing. Values are mean $\pm$ std over 3 runs. Best values for each score are bolded.

City	Model	LIV	PHY	NUI	SOC	AME	HOU
EINDHOVEN	BASE	0.095 $\pm$ 0.002	0.025 $\pm$ 0.001	0.059 $\pm$ 0.001	0.030 $\pm$ 0.001	0.054 $\pm$ 0.002	0.028$\pm$0.001
	AEF	0.089 $\pm$ 0.001	0.022$\pm$0.000	0.056 $\pm$ 0.001	0.028$\pm$0.000	0.045 $\pm$ 0.001	0.028$\pm$0.000
	AS	0.096 $\pm$ 0.000	0.027 $\pm$ 0.001	0.061 $\pm$ 0.001	0.030 $\pm$ 0.000	0.052 $\pm$ 0.001	0.029 $\pm$ 0.000
	TM	0.090 $\pm$ 0.002	0.024 $\pm$ 0.000	0.058 $\pm$ 0.001	0.030 $\pm$ 0.000	0.050 $\pm$ 0.001	0.028$\pm$0.000
	AEF+AS	0.090 $\pm$ 0.002	0.023 $\pm$ 0.000	0.056 $\pm$ 0.001	0.028$\pm$0.000	0.043 $\pm$ 0.001	0.029 $\pm$ 0.001
	AEF+TM	0.088$\pm$0.001	0.023 $\pm$ 0.001	0.055$\pm$0.001	0.028$\pm$0.000	0.044 $\pm$ 0.001	0.028$\pm$0.000
	AS+TM	0.091 $\pm$ 0.002	0.025 $\pm$ 0.000	0.059 $\pm$ 0.001	0.030 $\pm$ 0.001	0.049 $\pm$ 0.002	0.028$\pm$0.001
	AEF+AS+TM	0.090 $\pm$ 0.002	0.023 $\pm$ 0.000	0.057 $\pm$ 0.001	0.028$\pm$0.001	0.042$\pm$0.001	0.028$\pm$0.001
HENGEL	BASE	0.119 $\pm$ 0.001	0.029 $\pm$ 0.000	0.058 $\pm$ 0.003	0.033 $\pm$ 0.000	0.054 $\pm$ 0.002	0.040 $\pm$ 0.001
	AEF	0.107$\pm$0.001	0.026 $\pm$ 0.000	0.056$\pm$0.000	0.030$\pm$0.000	0.046 $\pm$ 0.000	0.039$\pm$0.001
	AS	0.113 $\pm$ 0.001	0.028 $\pm$ 0.000	0.060 $\pm$ 0.001	0.032 $\pm$ 0.000	0.050 $\pm$ 0.000	0.040 $\pm$ 0.000
	TM	0.112 $\pm$ 0.002	0.027 $\pm$ 0.000	0.059 $\pm$ 0.001	0.031 $\pm$ 0.001	0.049 $\pm$ 0.001	0.040 $\pm$ 0.000
	AEF+AS	0.111 $\pm$ 0.001	0.026 $\pm$ 0.001	0.059 $\pm$ 0.000	0.030$\pm$0.000	0.045 $\pm$ 0.002	0.040 $\pm$ 0.001
	AEF+TM	0.111 $\pm$ 0.001	0.026 $\pm$ 0.000	0.058 $\pm$ 0.000	0.030$\pm$0.001	0.043$\pm$0.002	0.041 $\pm$ 0.001
	AS+TM	0.111 $\pm$ 0.001	0.027 $\pm$ 0.000	0.060 $\pm$ 0.002	0.032 $\pm$ 0.000	0.047 $\pm$ 0.002	0.039$\pm$0.000
	AEF+AS+TM	0.110 $\pm$ 0.001	0.025$\pm$0.000	0.058 $\pm$ 0.001	0.030$\pm$0.001	0.043$\pm$0.001	0.039$\pm$0.001
DORDRECHT	BASE	0.114 $\pm$ 0.004	0.028 $\pm$ 0.000	0.068 $\pm$ 0.001	0.034 $\pm$ 0.000	0.060 $\pm$ 0.002	0.039 $\pm$ 0.001
	AEF	0.106 $\pm$ 0.003	0.026 $\pm$ 0.001	0.068 $\pm$ 0.001	0.034 $\pm$ 0.001	0.052 $\pm$ 0.002	0.036$\pm$0.000
	AS	0.111 $\pm$ 0.001	0.028 $\pm$ 0.001	0.066 $\pm$ 0.001	0.033$\pm$0.001	0.056 $\pm$ 0.001	0.039 $\pm$ 0.001
	TM	0.110 $\pm$ 0.001	0.028 $\pm$ 0.000	0.067 $\pm$ 0.001	0.034 $\pm$ 0.001	0.060 $\pm$ 0.000	0.037 $\pm$ 0.000
	AEF+AS	0.109 $\pm$ 0.000	0.026 $\pm$ 0.001	0.067 $\pm$ 0.001	0.033$\pm$0.000	0.048$\pm$0.001	0.037 $\pm$ 0.000
	AEF+TM	0.105$\pm$0.002	0.026 $\pm$ 0.001	0.068 $\pm$ 0.002	0.034 $\pm$ 0.000	0.051 $\pm$ 0.000	0.036$\pm$0.000
	AS+TM	0.110 $\pm$ 0.001	0.026 $\pm$ 0.000	0.065$\pm$0.002	0.033$\pm$0.001	0.055 $\pm$ 0.001	0.038 $\pm$ 0.001
	AEF+AS+TM	0.105$\pm$0.001	0.025$\pm$0.001	0.066 $\pm$ 0.001	0.033$\pm$0.000	0.049 $\pm$ 0.001	0.037 $\pm$ 0.001
BEESEL	BASE	0.078 $\pm$ 0.002	0.031 $\pm$ 0.002	0.037$\pm$0.003	0.028 $\pm$ 0.001	0.037 $\pm$ 0.003	0.022 $\pm$ 0.001
	AEF	0.082 $\pm$ 0.006	0.028$\pm$0.001	0.041 $\pm$ 0.005	0.026 $\pm$ 0.001	0.034 $\pm$ 0.005	0.021 $\pm$ 0.003
	AS	0.085 $\pm$ 0.007	0.035 $\pm$ 0.001	0.043 $\pm$ 0.005	0.028 $\pm$ 0.000	0.036 $\pm$ 0.002	0.024 $\pm$ 0.002
	TM	0.080 $\pm$ 0.007	0.035 $\pm$ 0.002	0.040 $\pm$ 0.005	0.027 $\pm$ 0.001	0.032$\pm$0.001	0.022 $\pm$ 0.003
	AEF+AS	0.081 $\pm$ 0.009	0.028$\pm$0.001	0.039 $\pm$ 0.004	0.024$\pm$0.001	0.035 $\pm$ 0.004	0.020 $\pm$ 0.001
	AEF+TM	0.077 $\pm$ 0.002	0.028$\pm$0.001	0.039 $\pm$ 0.002	0.025 $\pm$ 0.000	0.034 $\pm$ 0.003	0.019$\pm$0.001
	AS+TM	0.080 $\pm$ 0.005	0.035 $\pm$ 0.001	0.042 $\pm$ 0.002	0.026 $\pm$ 0.000	0.033 $\pm$ 0.002	0.022 $\pm$ 0.001
	AEF+AS+TM	0.073$\pm$0.002	0.029 $\pm$ 0.001	0.038 $\pm$ 0.001	0.025 $\pm$ 0.001	0.036 $\pm$ 0.004	0.019$\pm$0.001

5.2 Results on replacing modalities

In Table 4, we examine the performance when we exclude RS, DSM, NLRs, POI data, or all of the above during training. The goal is to see whether embeddings could mitigate performance degradation when important data sources are missing. Across all single-modality ablations, using any embeddings consistently improves the performance relative to the baseline model. Both AEF and TM usually have the largest improvements on BASE, indicating that AlphaEarth and TerraMind captures a lot of information related to urban livability. When the input data excludes DSM, NLRs, and POI, AEF recovers similar performance to the full-data models trained in Table 1. The RS ablation is slightly more difficult: AEF does improve upon BASE in this setting, but it still has worse LIV RMSE than the full-data AEF

Table 3: RMSE comparison across livability domain scores between baseline network and models that include foundation model embeddings. The POI column indicates where there is POI data or not. Values are mean $\pm$ std over 3 runs. Best values for each score are bolded.

POI	Model	LIV	PHY	NUI	SOC	AME	HOU
NULL	BASE	0.093 $\pm$ 0.001	0.031 $\pm$ 0.000	0.050 $\pm$ 0.000	0.034 $\pm$ 0.000	0.048 $\pm$ 0.001	0.031 $\pm$ 0.000
	AEF	0.089 $\pm$ 0.001	0.028$\pm$0.001	0.050 $\pm$ 0.001	0.032 $\pm$ 0.000	0.041 $\pm$ 0.001	0.031 $\pm$ 0.000
	AS	0.095 $\pm$ 0.003	0.033 $\pm$ 0.000	0.053 $\pm$ 0.002	0.034 $\pm$ 0.000	0.045 $\pm$ 0.002	0.032 $\pm$ 0.000
	TM	0.089 $\pm$ 0.002	0.031 $\pm$ 0.001	0.049 $\pm$ 0.001	0.033 $\pm$ 0.000	0.045 $\pm$ 0.000	0.031 $\pm$ 0.001
	AEF+AS	0.090 $\pm$ 0.002	0.029 $\pm$ 0.001	0.048 $\pm$ 0.001	0.032 $\pm$ 0.000	0.041 $\pm$ 0.001	0.030$\pm$0.001
	AEF+TM	0.086$\pm$0.000	0.029 $\pm$ 0.000	0.048 $\pm$ 0.002	0.032 $\pm$ 0.000	0.040$\pm$0.001	0.031 $\pm$ 0.000
	AS+TM	0.090 $\pm$ 0.002	0.032 $\pm$ 0.001	0.050 $\pm$ 0.001	0.033 $\pm$ 0.001	0.043 $\pm$ 0.002	0.032 $\pm$ 0.000
	AEF+AS+TM	0.087 $\pm$ 0.001	0.028$\pm$0.000	0.047$\pm$0.001	0.031$\pm$0.000	0.040$\pm$0.001	0.031 $\pm$ 0.000
ANY	BASE	0.108 $\pm$ 0.002	0.026 $\pm$ 0.001	0.062 $\pm$ 0.001	0.032 $\pm$ 0.001	0.056 $\pm$ 0.002	0.035 $\pm$ 0.000
	AEF	0.099$\pm$0.001	0.024 $\pm$ 0.000	0.060$\pm$0.000	0.030 $\pm$ 0.000	0.048 $\pm$ 0.001	0.033$\pm$0.000
	AS	0.105 $\pm$ 0.000	0.027 $\pm$ 0.001	0.063 $\pm$ 0.001	0.031 $\pm$ 0.000	0.053 $\pm$ 0.000	0.034 $\pm$ 0.000
	TM	0.102 $\pm$ 0.001	0.025 $\pm$ 0.000	0.061 $\pm$ 0.000	0.031 $\pm$ 0.000	0.053 $\pm$ 0.001	0.034 $\pm$ 0.000
	AEF+AS	0.101 $\pm$ 0.001	0.024 $\pm$ 0.000	0.061 $\pm$ 0.001	0.029$\pm$0.000	0.045$\pm$0.001	0.034 $\pm$ 0.000
	AEF+TM	0.099$\pm$0.000	0.024 $\pm$ 0.001	0.060$\pm$0.000	0.030 $\pm$ 0.000	0.046 $\pm$ 0.001	0.034 $\pm$ 0.001
	AS+TM	0.102 $\pm$ 0.002	0.026 $\pm$ 0.000	0.062 $\pm$ 0.001	0.031 $\pm$ 0.001	0.051 $\pm$ 0.002	0.034 $\pm$ 0.000
	AEF+AS+TM	0.100 $\pm$ 0.001	0.023$\pm$0.000	0.061 $\pm$ 0.000	0.029$\pm$0.000	0.045$\pm$0.001	0.034 $\pm$ 0.001

network. This reflects the fine-grained information not captured by embeddings but reflected in RS imagery.

We further examine if the model can solely learn from combinations of embedding without RS, DSM, NLRs, and POI data. Out of these, the highest performing model is AEF+TM, and it achieves similar scores to the full-data BASE model on all metrics and even outperforms some domains like PHY and AME. Incorporating AnySat into AEF+TM hurts RMSE, so AnySat may introduce redundant or noisy information without other modalities.

5.3 Results on missing modality

We also examine how the trained full-data models in Table 1 respond when one of their inputs is missing to understand how models weigh different modalities. In Table 5, we set one of the modalities to zero and calculate the resulting RMSE. Models that include AlphaEarth tend to be more robust towards this ablation, so these embeddings would most effectively compensate for missing modalities. In particular, AEF+AS+TM tends to be robust with missing modalities as it likely places more weight on the foundation model embeddings. The largest performance gap between the baseline and augmented models occur when RS is missing. This indicates that RS carries the most unique information, which is consistent

Table 4: RMSE comparison across livability domain scores between baseline network and models that include foundation model embeddings. The ablation column indicates the modality excluded during training. Values are mean $\pm$ std over 3 runs. Best values for each score are bolded.

Ablation	Model	LIV	PHY	NUI	SOC	AME	HOU
RS	BASE	0.122 $\pm$ 0.000	0.029 $\pm$ 0.000	0.067 $\pm$ 0.001	0.036 $\pm$ 0.000	0.059 $\pm$ 0.002	0.038 $\pm$ 0.000
	AEF	0.105$\pm$0.001	0.024$\pm$0.000	0.062$\pm$0.000	0.032$\pm$0.000	0.047$\pm$0.002	0.035$\pm$0.000
	AS	0.110 $\pm$ 0.001	0.028 $\pm$ 0.001	0.065 $\pm$ 0.001	0.034 $\pm$ 0.000	0.055 $\pm$ 0.001	0.037 $\pm$ 0.000
	TM	0.105$\pm$0.002	0.026 $\pm$ 0.000	0.062$\pm$0.001	0.032$\pm$0.000	0.054 $\pm$ 0.000	0.035$\pm$0.000
DSM	BASE	0.104 $\pm$ 0.002	0.028 $\pm$ 0.001	0.062 $\pm$ 0.000	0.032 $\pm$ 0.000	0.057 $\pm$ 0.000	0.034 $\pm$ 0.000
	AEF	0.100$\pm$0.002	0.024$\pm$0.000	0.060$\pm$0.001	0.031$\pm$0.000	0.048$\pm$0.001	0.033$\pm$0.001
	AS	0.103 $\pm$ 0.000	0.028 $\pm$ 0.001	0.062 $\pm$ 0.000	0.031$\pm$0.001	0.054 $\pm$ 0.001	0.033$\pm$0.000
	TM	0.100$\pm$0.001	0.027 $\pm$ 0.000	0.060$\pm$0.001	0.031$\pm$0.000	0.052 $\pm$ 0.000	0.033$\pm$0.000
NLRS	BASE	0.108 $\pm$ 0.001	0.029 $\pm$ 0.000	0.063 $\pm$ 0.001	0.033 $\pm$ 0.000	0.059 $\pm$ 0.001	0.034$\pm$0.001
	AEF	0.101$\pm$0.003	0.025$\pm$0.000	0.061$\pm$0.001	0.031$\pm$0.001	0.049$\pm$0.002	0.034$\pm$0.001
	AS	0.107 $\pm$ 0.003	0.028 $\pm$ 0.001	0.063 $\pm$ 0.001	0.032 $\pm$ 0.000	0.057 $\pm$ 0.002	0.035 $\pm$ 0.000
	TM	0.102 $\pm$ 0.002	0.027 $\pm$ 0.000	0.061$\pm$0.002	0.031$\pm$0.000	0.053 $\pm$ 0.001	0.034$\pm$0.000
POI	BASE	0.111 $\pm$ 0.017	0.027 $\pm$ 0.001	0.067 $\pm$ 0.010	0.037 $\pm$ 0.010	0.062 $\pm$ 0.003	0.037 $\pm$ 0.005
	AEF	0.099 $\pm$ 0.002	0.025$\pm$0.000	0.060 $\pm$ 0.001	0.030$\pm$0.000	0.048$\pm$0.002	0.034 $\pm$ 0.000
	AS	0.103 $\pm$ 0.002	0.028 $\pm$ 0.000	0.062 $\pm$ 0.001	0.032 $\pm$ 0.000	0.054 $\pm$ 0.002	0.034 $\pm$ 0.000
	TM	0.098$\pm$0.001	0.026 $\pm$ 0.000	0.059$\pm$0.000	0.030$\pm$0.001	0.056 $\pm$ 0.001	0.033$\pm$0.000
RS	AEF	0.112 $\pm$ 0.001	0.024$\pm$0.001	0.065 $\pm$ 0.001	0.034 $\pm$ 0.000	0.051 $\pm$ 0.002	0.036 $\pm$ 0.001
	AS	0.115 $\pm$ 0.002	0.027 $\pm$ 0.001	0.067 $\pm$ 0.001	0.035 $\pm$ 0.000	0.064 $\pm$ 0.001	0.037 $\pm$ 0.000
	TM	0.109 $\pm$ 0.001	0.025 $\pm$ 0.000	0.063 $\pm$ 0.001	0.033 $\pm$ 0.000	0.056 $\pm$ 0.001	0.035$\pm$0.000
	AEF+AS	0.113 $\pm$ 0.001	0.024$\pm$0.000	0.065 $\pm$ 0.001	0.033 $\pm$ 0.000	0.050$\pm$0.003	0.036 $\pm$ 0.000
DSM	AEF+TM	0.107$\pm$0.001	0.024$\pm$0.000	0.062$\pm$0.001	0.032$\pm$0.000	0.050$\pm$0.002	0.036 $\pm$ 0.000
	AS+TM	0.109 $\pm$ 0.002	0.026 $\pm$ 0.000	0.066 $\pm$ 0.001	0.034 $\pm$ 0.000	0.059 $\pm$ 0.003	0.036 $\pm$ 0.000
	AEF+AS+TM	0.110 $\pm$ 0.003	0.026 $\pm$ 0.002	0.063 $\pm$ 0.001	0.033 $\pm$ 0.003	0.050$\pm$0.005	0.036 $\pm$ 0.002

with related studies [34]. Notably, zeroing out POI yields a counterintuitive result: BASE achieves a LIV RMSE of 0.099, lower than its full-data score of 0.106 (Table 1). This is consistent with the POI stratification results in Table 3 (single-seed), where BASE models perform worse in areas with POI present (0.105) compared to areas without POI (0.092). Together, these results suggest that POI data introduces heterogeneity that complicates overall livability prediction, even while improving domain-specific scores like AME, likely because dense POI areas are more spatially variable and harder to predict uniformly.

5.4 Geographic Variation

Illustrated in Figure 3, we examine how the full-data BASE and AEF predictions differ spatially across four test cities. In the right-most column, we plot the difference between the LIV L1 scores for both BASE and AEF. Orange indicates regions where AEF has lower L1 than BASE, and purple indicates where BASE performs better. In Eindhoven and Hengelo, the difference maps show small but consistent orange regions, which illustrates that AlphaEarth improves predictions across most urban areas. Dordrecht shows a mixed pattern, with purple concentrated near the historical center. In contrast, Beesel is predominantly purple, confirming that AEF underperforms BASE in this rural community, consistent with the quantitative results in Table 2.

6 Discussion

This following section consists of explainability techniques to understand how the model weighs AlphaEarth versus other modalities. As such, we analyze the Transformer attention mechanism and its entropy structure, as well as gradient-based activation maps within the convolutional branches.

6.1 Attention

Transformers use attention to weigh the relative importance of different modalities. We can use this to gain insight into how the

Table 5: RMSE comparison across livability domain scores between baseline network and models that include foundation model embeddings. All models are trained on RS, DSM, NLRS, POI, and the specified embeddings. The ablation column indicates what modality was zeroed out during testing. Values are mean $\pm$ std over 3 runs. Best values for each score are bolded.

Ablation	Model	LIV	PHY	NUI	SOC	AME	HOU
RS	BASE	0.156 $\pm$ 0.007	0.039 $\pm$ 0.004	0.076 $\pm$ 0.002	0.049 $\pm$ 0.003	0.068 $\pm$ 0.004	0.050 $\pm$ 0.007
	AEF	0.112 $\pm$ 0.003	0.026 $\pm$ 0.000	0.063 $\pm$ 0.002	0.034 $\pm$ 0.001	0.048 $\pm$ 0.001	0.036 $\pm$ 0.001
	AS	0.138 $\pm$ 0.006	0.038 $\pm$ 0.003	0.074 $\pm$ 0.003	0.044 $\pm$ 0.003	0.066 $\pm$ 0.002	0.041 $\pm$ 0.001
	TM	0.117 $\pm$ 0.002	0.030 $\pm$ 0.001	0.066 $\pm$ 0.001	0.039 $\pm$ 0.002	0.063 $\pm$ 0.001	0.038 $\pm$ 0.001
	AEF+AS	0.111 $\pm$ 0.002	0.026 $\pm$ 0.001	0.062 $\pm$ 0.001	0.033 $\pm$ 0.001	0.045 $\pm$ 0.001	0.036 $\pm$ 0.001
	AEF+TM	0.108$\pm$0.001	0.026 $\pm$ 0.001	0.061$\pm$0.001	0.033 $\pm$ 0.000	0.046 $\pm$ 0.001	0.036 $\pm$ 0.001
	AS+TM	0.119 $\pm$ 0.010	0.031 $\pm$ 0.003	0.069 $\pm$ 0.005	0.040 $\pm$ 0.005	0.063 $\pm$ 0.007	0.039 $\pm$ 0.003
	AEF+AS+TM	0.108$\pm$0.001	0.025$\pm$0.001	0.062 $\pm$ 0.001	0.032$\pm$0.001	0.044$\pm$0.001	0.035$\pm$0.000
DSM	BASE	0.107 $\pm$ 0.002	0.033 $\pm$ 0.002	0.061 $\pm$ 0.001	0.032 $\pm$ 0.001	0.059 $\pm$ 0.001	0.034 $\pm$ 0.001
	AEF	0.098$\pm$0.001	0.026 $\pm$ 0.001	0.059$\pm$0.000	0.030$\pm$0.000	0.047 $\pm$ 0.001	0.033$\pm$0.000
	AS	0.103 $\pm$ 0.001	0.030 $\pm$ 0.001	0.062 $\pm$ 0.002	0.032 $\pm$ 0.000	0.056 $\pm$ 0.001	0.033$\pm$0.000
	TM	0.101 $\pm$ 0.001	0.029 $\pm$ 0.000	0.060 $\pm$ 0.000	0.031 $\pm$ 0.000	0.055 $\pm$ 0.001	0.034 $\pm$ 0.000
	AEF+AS	0.100 $\pm$ 0.001	0.026 $\pm$ 0.001	0.060 $\pm$ 0.001	0.030$\pm$0.000	0.045 $\pm$ 0.000	0.033$\pm$0.000
	AEF+TM	0.098$\pm$0.001	0.026 $\pm$ 0.001	0.059$\pm$0.001	0.030$\pm$0.000	0.046 $\pm$ 0.001	0.033$\pm$0.000
	AS+TM	0.101 $\pm$ 0.002	0.029 $\pm$ 0.001	0.061 $\pm$ 0.002	0.031 $\pm$ 0.000	0.054 $\pm$ 0.001	0.033$\pm$0.001
	AEF+AS+TM	0.099 $\pm$ 0.001	0.025$\pm$0.000	0.060 $\pm$ 0.000	0.030$\pm$0.000	0.044$\pm$0.001	0.033$\pm$0.001
NLRS	BASE	0.111 $\pm$ 0.002	0.039 $\pm$ 0.001	0.064 $\pm$ 0.001	0.033$\pm$0.000	0.070 $\pm$ 0.002	0.035 $\pm$ 0.000
	AEF	0.104$\pm$0.002	0.027 $\pm$ 0.001	0.061$\pm$0.000	0.033$\pm$0.001	0.048 $\pm$ 0.001	0.034$\pm$0.000
	AS	0.107 $\pm$ 0.002	0.038 $\pm$ 0.002	0.064 $\pm$ 0.002	0.033$\pm$0.000	0.063 $\pm$ 0.003	0.035 $\pm$ 0.000
	TM	0.109 $\pm$ 0.002	0.035 $\pm$ 0.000	0.062 $\pm$ 0.001	0.033$\pm$0.000	0.063 $\pm$ 0.004	0.035 $\pm$ 0.000
	AEF+AS	0.105 $\pm$ 0.001	0.027 $\pm$ 0.001	0.062 $\pm$ 0.001	0.033$\pm$0.001	0.046 $\pm$ 0.000	0.035 $\pm$ 0.000
	AEF+TM	0.104$\pm$0.002	0.028 $\pm$ 0.001	0.062 $\pm$ 0.001	0.033$\pm$0.001	0.046 $\pm$ 0.001	0.034$\pm$0.001
	AS+TM	0.107 $\pm$ 0.001	0.035 $\pm$ 0.002	0.062 $\pm$ 0.001	0.033$\pm$0.001	0.059 $\pm$ 0.004	0.035 $\pm$ 0.000
	AEF+AS+TM	0.104$\pm$0.001	0.026$\pm$0.002	0.062 $\pm$ 0.001	0.033$\pm$0.000	0.045$\pm$0.001	0.034$\pm$0.000
POI	BASE	0.099 $\pm$ 0.002	0.027 $\pm$ 0.000	0.063 $\pm$ 0.001	0.033 $\pm$ 0.000	0.069 $\pm$ 0.003	0.034 $\pm$ 0.000
	AEF	0.098 $\pm$ 0.001	0.025$\pm$0.001	0.061 $\pm$ 0.000	0.031 $\pm$ 0.001	0.053 $\pm$ 0.001	0.034 $\pm$ 0.000
	AS	0.099 $\pm$ 0.001	0.027 $\pm$ 0.000	0.063 $\pm$ 0.001	0.032 $\pm$ 0.000	0.065 $\pm$ 0.003	0.034 $\pm$ 0.000
	TM	0.097 $\pm$ 0.002	0.026 $\pm$ 0.001	0.062 $\pm$ 0.002	0.032 $\pm$ 0.000	0.058 $\pm$ 0.001	0.034 $\pm$ 0.000
	AEF+AS	0.099 $\pm$ 0.001	0.025$\pm$0.000	0.061 $\pm$ 0.002	0.030 $\pm$ 0.000	0.049 $\pm$ 0.001	0.033$\pm$0.000
	AEF+TM	0.096$\pm$0.002	0.026 $\pm$ 0.001	0.059$\pm$0.001	0.030 $\pm$ 0.001	0.050 $\pm$ 0.003	0.034 $\pm$ 0.001
	AS+TM	0.096$\pm$0.000	0.027 $\pm$ 0.001	0.062 $\pm$ 0.001	0.032 $\pm$ 0.000	0.055 $\pm$ 0.004	0.034 $\pm$ 0.000
	AEF+AS+TM	0.098 $\pm$ 0.001	0.025$\pm$0.000	0.061 $\pm$ 0.001	0.029$\pm$0.001	0.048$\pm$0.002	0.034 $\pm$ 0.000

model processes AlphaEarth embeddings. Notably, high attention can have two interpretations: either the modality is crucial, or there is significant heterogeneity within the modality. In the latter case, the model needs to learn the internal differences within that modality, causing the transformer to assign high attention. With this in mind, Figure 4 displays average attention heatmaps for BASE and AEF models in 2 distinct groups: high and low livability areas.

In the figure, the RS and POI attention scores both decrease when AlphaEarth is included, while DSM and NLRS attention scores increase. One interpretation is that AlphaEarth encodes information that overlaps with RS and POI, leading the model to redistribute attention away from those modalities. However, attention weights alone do not establish this as high attention may reflect internal heterogeneity. Consistent with this caution, the missing-modality results in Table 5 show that AEF substantially reduces the RMSE penalty when RS is zeroed out (from 0.156 to 0.112 for LIV), lending indirect support to the view that AlphaEarth partially compensates for missing RS information.

Both BASE and AEF assign POI to have the highest attention of any modality in Figure 4, even though POI gives less information for livability prediction. This is backed up since the performance drop when we zero-out POI isn't as significant as RS or NLRS in Table 5. Consistent with the explanation proposed for the same phenomenon in [34], POI draws the most attention because the model must learn to resolve substantial internal variation in the text from empty strings to dense, heterogeneous facility listings. In addition, POI is the modality most directly related to AME, so part of POI's attention may reflect the model being important for one of the domains rather than broadly important across all outputs.

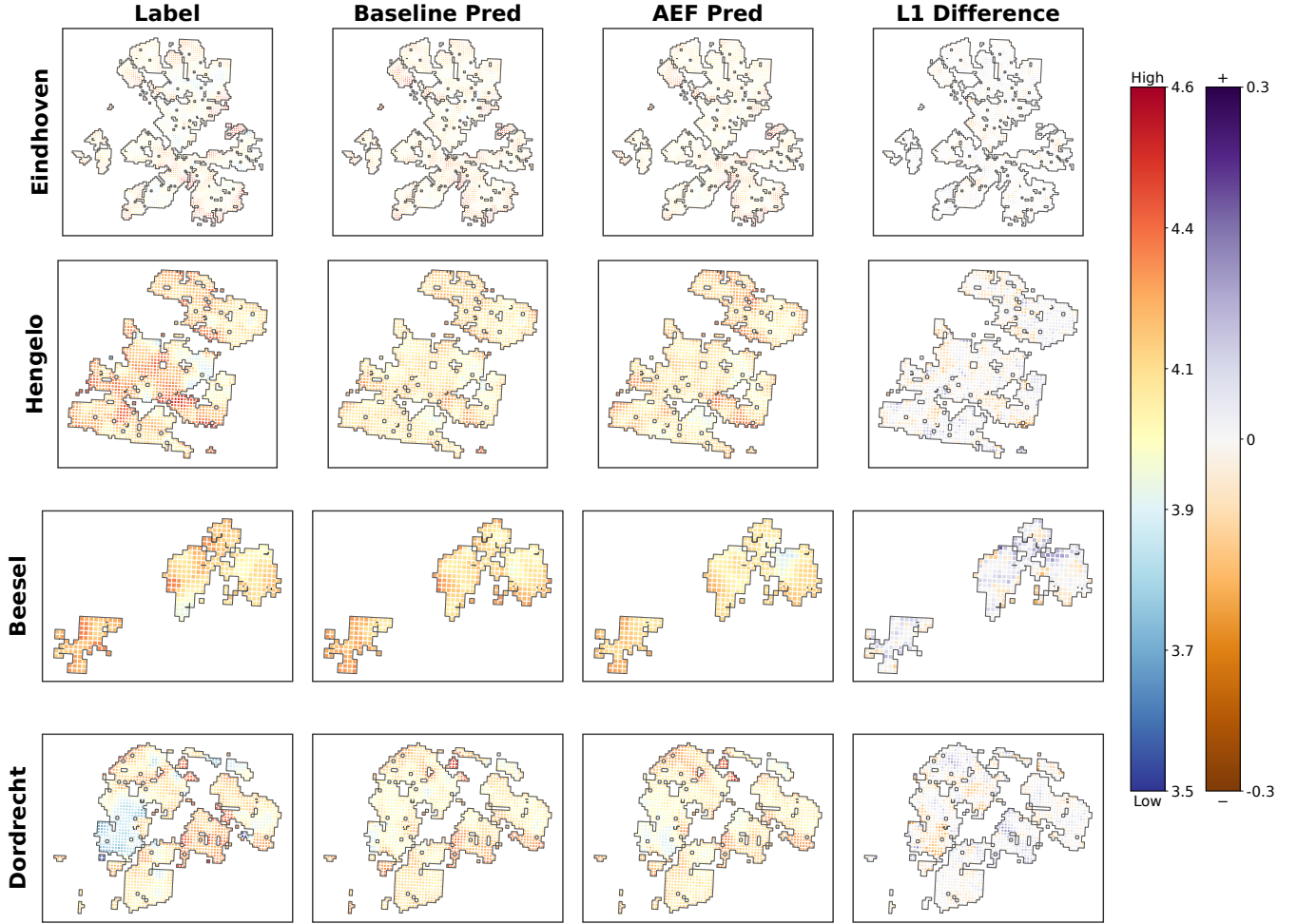


Figure 3: The overall livability score map of several test cities: Eindhoven, Hengelo, Beesel, and Dordrecht. From left to right, we present: the ground truth label, the BASE prediction, the AEF prediction, and the difference between the L1 scores of the BASE and AEF models. The left three columns correspond to the leftmost colorbar. The rightmost column corresponds to the rightmost colorbar, where higher scores indicate AEF has a higher L1 than BASE.

To understand how the attention is used differently between the BASE and AEF models, we compute the entropy of the attention distribution across modalities for each sample. High-livability areas exhibit significantly higher attention entropy in both the BASE and AEF models, indicating that attention is more broadly distributed across modalities rather than concentrated. This entropy gap is small in absolute terms but consistent across both models (BASE +0.012, 95% CI [0.008, 0.016], $p = 1.65 \times 10^{-10}$; AEF +0.022, 95% CI [0.019, 0.026], $p = 1.71 \times 10^{-32}$). Furthermore, AEF attention weights are positively correlated with both LIV scores (Pearson $r = 0.25$, 95% CI [0.23, 0.26], $p = 2.27 \times 10^{-186}$) and attention entropy ($r = 0.26$, 95% CI [0.25, 0.28], $p = 1.69 \times 10^{-209}$), suggesting that high-livability areas exhibit greater input heterogeneity across spatial modalities, prompting the model to draw on a broader set of embeddings rather than relying more heavily on AEF specifically. In low-livability areas, POI dominates attention in both models, whereas high-livability areas distribute attention more evenly across RS, NLRS, and AEF.

6.2 Full Grad-CAM

We apply Full Grad-CAM to understand how the inclusion of AlphaEarth changes the spatial attention of the model across the RS, DSM, and NLRS branches. Gradient-weighted Class Activation Mapping (Grad-CAM) [26] is an explainability technique to describe the decision-making process of convolutional neural networks. It involves using the gradient of the specified target concept to produce a coarse localization map, highlighting the most significant areas in the image. Standard Grad-CAM applies a ReLU to the weighted activation maps, only allowing positive contributions to the target output. We instead use Full Grad-CAM, which removes this ReLU operation, allowing both positive and negative feature contributions to be visualized. The core form for Full Grad-CAM is:

$$L(x, y) = \sum_k \alpha_k^c A^k(x, y) \quad (2)$$

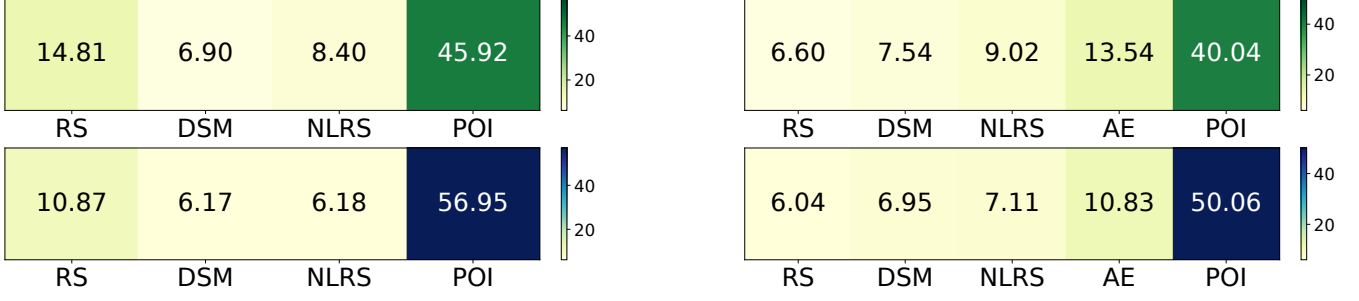


Figure 4: Average attention weights for the top and bottom 50% of samples by true overall livability scores. The top and bottom rows correspond to the high and low livability groups, respectively. The left and right heatmaps correspondingly refer to the BASE and AEF models. AE corresponds to AlphaEarth embeddings.

where $A^k(x, y)$ is the activation value of the k -th convolutional feature map at spatial location (x, y) , and α_k^c represents the gradient-based weight for the k -th feature map with respect to the c -th regression output.

In Figure 5, we examine how Full Grad-CAM differs between the BASE and AEF models on a test sample in the city of Dordrecht. The inclusion of AlphaEarth produces notable changes in how the model attends to specific regions across all three modalities, which we highlight with three annotated examples. In particular, the section highlighted ① is a park. The AEF model has a much stronger relative Grad-CAM score in this zone, suggesting that the inclusion of global embeddings enables the model to identify structures near the image edge. In DSM, the ② region, corresponding to a residential home, has a much stronger positive Grad-CAM response in the BASE model. This is inconsistent with the RS Grad-CAM where both models already assigned the same region with a negative contribution. The AEF model resolves this inconsistency, suppressing the positive DSM response and aligning it with the RS signal, suggesting that AlphaEarth helps the model integrate information more consistently across modalities. The final area denoted by ③ is an example of how the AEF model is able to attend to regions near the border in NLRs. This particular section corresponds to an old water tower that has been turned into a landmark with surrounding walkable gardens. AlphaEarth’s pretraining on diverse geospatial data may encode semantic knowledge of such landmarks, enabling the model to assign higher importance to this region in NLRs despite its proximity to the image edge. Across all three regions, the inclusion of AlphaEarth enables the model to better integrate spatial context across modalities and attend to semantically meaningful structures near the image boundary.

7 Conclusion

In this study, we have demonstrated an effective approach to incorporating geospatial foundation model embeddings into urban livability evaluation. We introduce a way to include geospatial foundation model embeddings through convolutions in a Transformer model. From this, we found that AlphaEarth and other embeddings can improve the performance of the original livability evaluation model and substantially mitigate performance degradation when individual modalities such as RS or POI are unavailable. However,

AlphaEarth does not fully substitute for these modalities: when RS is excluded, the AEF model still underperforms the full-data AEF network, reflecting fine-grained spatial information present in high-resolution imagery that embeddings do not completely capture.

Based on our results, we offer the following guidance for practitioners deploying this framework in new contexts. AlphaEarth consistently provides the most reliable performance gains and is a suitable embedding choice when operating under a single-embedding budget. In predominantly urban contexts, AEF alone or AEF+TM offers the best cost-benefit tradeoff. However, practitioners working in rural or mixed urban-rural regions should not rely on a single embedding — single embeddings like AEF or AS can worsen baseline performance in rural areas, and only multi-embedding combinations such as AEF+AS+TM reliably recover and improve upon the baseline. This likely reflects a distributional mismatch between the predominantly urban composition of embedding pretraining data and rural geospatial patterns, and should be treated as a deployment risk until embeddings with broader geographic coverage become available. When key modalities are unavailable, embeddings offer meaningful mitigation: AEF alone recovers near full-data performance when DSM, NLRs, or POI are missing, and AEF+TM approaches full-data baseline performance even without any of the four standard modalities. RS remains the hardest modality to substitute, as even the best embedding combinations do not fully close the performance gap when RS is absent.

We further examine interpretability through attention entropy analysis and find that the higher AEF attention weights observed in high-livability areas reflect greater input heterogeneity rather than increased reliance on AlphaEarth specifically — high-livability areas distribute attention more broadly across modalities, while low-livability areas are dominated by POI. We also plot Grad-CAM values and find that AlphaEarth embeddings can unify the Grad-CAM values of different modalities, identify important landmarks near the image edge, and increase the Grad-CAM values of NLRs. While the embeddings do help generalizability, its performance boost is still lower than expected. This suggests that the embeddings are primarily helpful in geographical areas of data scarcity. However, there are no established validation techniques for rural livability evaluation. In future work, we plan to examine how embeddings contribute to spatial biases and how these can be mitigated.

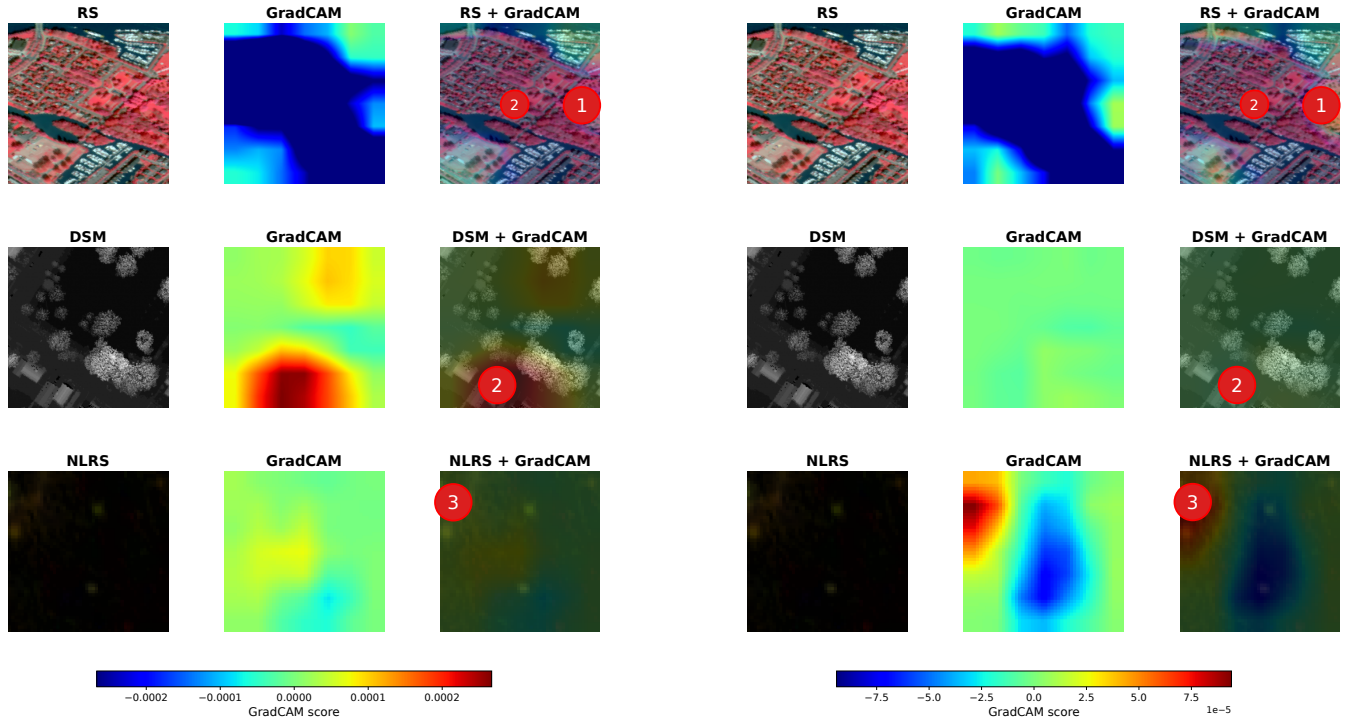


Figure 5: A sample for full Grad-CAM visualization for overall livability scores. The left and right Grad-CAM scores respectively correspond to the BASE and AEF models. The red circles correspond to important areas that affect livability, and they are explained in Section 6.2.

through architecture design. Extending the framework beyond the Netherlands to test cross-cultural and cross-urban generalization is another important direction. Finally, incorporating temporal embeddings is an interesting research direction to explore how livability evolves over time, opening the door to longitudinal urban analysis.

Limitations. All results in this paper are evaluated in the Netherlands, using the LBM dataset. Although we test on rural regions, we cannot yet validate whether these findings transfer to other regions that lack the ground-truth labels or the geospatial inputs necessary to train the model. We see this as a limitation and note it as an important area for future work. Applying this model to other livability or socioeconomic benchmarks outside the Netherlands would let us test whether the preference of AlphaEarth and multi-embedding combinations would hold on out-of-distribution areas that these embeddings were meant to be validated on.

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Ethics and Privacy Statement

This study uses only publicly available, non-personal data: satellite and elevation imagery, point-of-interest listings, geospatial foundation model embeddings, and the Dutch government’s aggregated Leefbaarometer livability statistics, all reported at the level of $100\text{ m} \times 100\text{ m}$ grid cells rather than individuals or households. No human-subjects data was collected, and no institutional review was required. The primary societal risk we identify is downstream misuse: livability predictions of this kind could be applied to real estate valuation, insurance underwriting, or resource-allocation decisions in ways that entrench existing inequality, particularly given the rural/urban performance disparities and embedding-driven biases we document. We intend this work as a research contribution to inform human-supervised policy and planning analysis, not as an input to automated eligibility or pricing decisions, and we report these disparities explicitly so that practitioners are aware of where the model is least reliable.

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A Training Details

All models were trained on a single NVIDIA H100 with 80 GB of HBM3 memory, utilizing CUDA 13.1 for GPU acceleration. The models were trained using PyTorch for 12 epochs with a batch size of 16. Our optimizer was AdamW [19] with a learning rate of $5e - 5$, weight decay of 0.1, epsilon of $1e - 8$, and maximum gradient norm of 1. We used early stopping with a patience of 5. We used BERT to create the text embedding from integer token indices with a sequence length of 512. DenseNet extracted RS, DSM, and NLRS image embeddings of shape 3×1024 . The convolution layers also extracted 3×1024 embeddings from AlphaEarth. The text

and image inputs were concatenated and fed into a Transformer Encoder with 12 self-attention heads, 768 hidden units, and an activation function of Gaussian Error Linear Unit (GELU) [9]. We set Dropout to 0.1, warmup steps to 0, and gradient accumulation steps to 1. Although the original paper [34] utilized MSE as the loss function, we use MAE because it had slightly better performance on a majority of outputs, as illustrated in Table 6. The table only recorded the performance for one model on the same seed.

Table 6: RMSE comparison between loss functions on base-line model across livability domain scores. Best values for each score are bolded.

Model	LIV	PHY	NUI	SOC	AME	HOU
MAE	0.104	0.026	0.061	0.031	0.054	0.034
MSE	0.104	0.027	0.061	0.032	0.053	0.034

Each embedding source (AlphaEarth, AnySat, TerraMind) is processed by its own dedicated convolutional branch; branches do not share weights with each other or with the DenseNet/BERT backbones. Each branch takes the raw embedding as a $C_{in} \times H \times W$ tensor ($C_{in} = 64$, $H=W=50$ for AlphaEarth; $C_{in} = 1536$, $H=W=24$ for AnySat; $C_{in} = 384$, $H=W=14$ for TerraMind) and applies three stacked 3×3 convolutions with stride 1 and padding 1, each mapping to 128 output channels and each followed by batch normalization and a ReLU activation. The resulting $128 \times H \times W$ feature map is

spatially reduced by adaptive average pooling to 3×1 locations, flattened, and linearly projected from 128 to 1024 dimensions, followed by a ReLU and dropout with rate 0.1, yielding the same 3×1024 token shape used by the DenseNet branches so all modalities can be concatenated before the Transformer Encoder.

B Comparison to Published TMTMR Baseline

Table 7: RMSE comparison between our reimplemented BASE model and the published TMTMR baseline [34] (BASE-MSE) across livability domain scores. Best values for each score are bolded.

Model	LIV	PHY	NUI	SOC	AME	HOU
BASE-MSE [34]	0.0966	0.0248	0.0580	0.0294	0.0488	0.0327
BASE	0.106±0.002	0.027±0.000	0.061±0.001	0.032±0.001	0.055±0.002	0.034±0.000

Table 7 compares our reimplemented BASE model against the published TMTMR baseline from Ref.[34], which we label BASE-MSE since it is trained with an MSE loss. BASE consistently trails BASE-MSE by roughly 5 to 11% RMSE across all six scores. We attribute this gap to differences in training we did not attempt to replicate, such as BASE-MSE being trained for 20 epochs instead of 12 or BASE-MSE using MSE loss instead of MAE. We selected MAE because it performed marginally better in our own environment (Table 6). Because of this gap, all embedding-related comparisons in this paper are made relative to our own BASE reimplementation rather than to BASE-MSE, which we report here for reference only.